

WINNER ENGINEERING LEARNING HUB SERIES #04

FUNDAMENTAL TOPIC

- ❖ **Maintenance Space Awareness, Access Panel (For all Equipments)**
- ❖ **Opening Requirement**
- ❖ **Types of different drawings, Do's & Don't Standard**
(Shop Drawing, CSD, Room Book, RCP, etc)

Table of Content

❖ Maintenance Space Awareness, Access Panel

Access panel requirement is depend on type of ceiling if equipment installs above ceiling.

1. Ceiling Cassette, Ducted Fan Coil Unit, Air Handling Unit
2. Fire Damper, Volume Control Damper, Non-Return Damper, VAV Box, Duct Heater
3. Cooling Tower

❖ Opening Requirement

Access panels can vary in size, design, and location depending on the specific requirements of the facility or system

1. 450 x 450, 600 x 600 (Standard Sizes)

❖ Types of different drawings, Do's & Don't Standard (Room Book, RCP, Setting Out, etc)

Why Maintenance Space Awareness, Access Panel needs?

- ▶ Means knowledge and understanding of the physical spaces within M&E systems that are designated for maintenance activities
- ▶ Specifically designed to provide convenient access to critical components, equipment, or systems that may require routine inspection, repair, or servicing
- ▶ Maintenance personnel to access equipment and systems easily, perform inspections and repairs promptly, and ensure that the facility or system remains in optimal working condition
- ▶ Having proper maintenance space awareness and well-designed access panels is crucial for effective and efficient maintenance operations

Maintenance Space Awareness, Access Panel

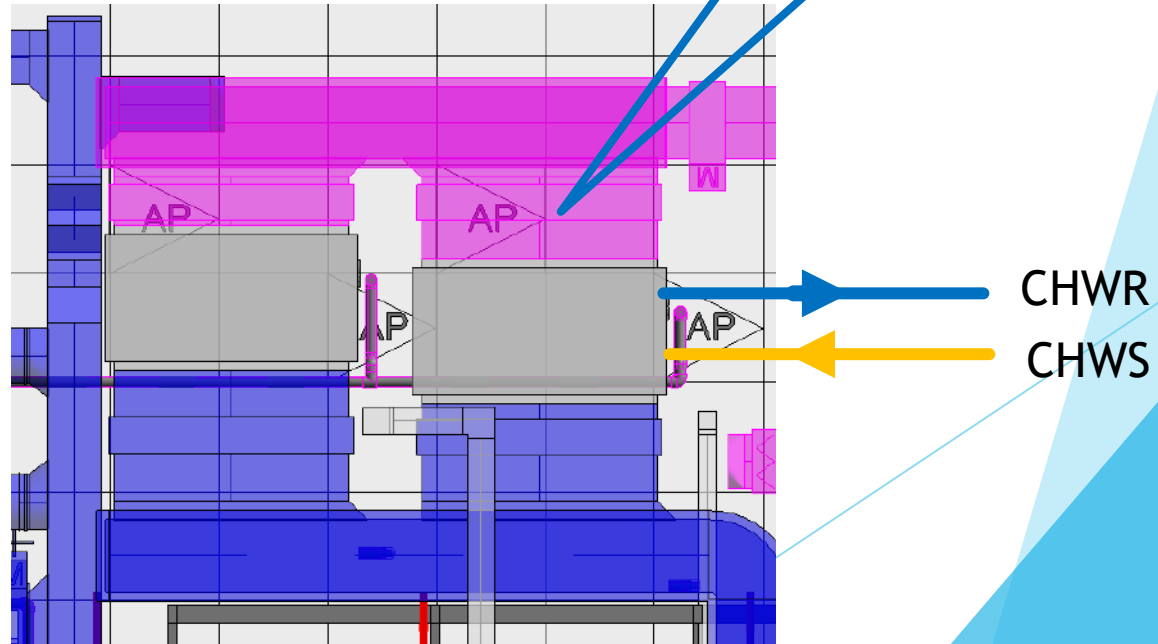


Ducted Fan Coil Unit

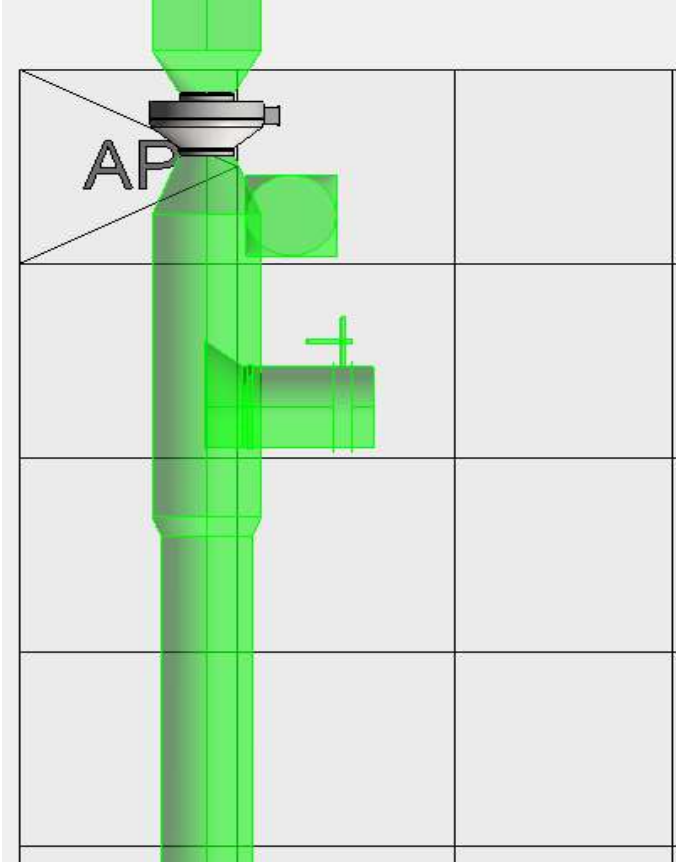
Ceiling Cassette

- Maintenance and Repairs
- Filter Replacement
- Electrical Connections
- System Adjustments
- Troubleshooting

- Near RA plenum box
- CHW pipes connection

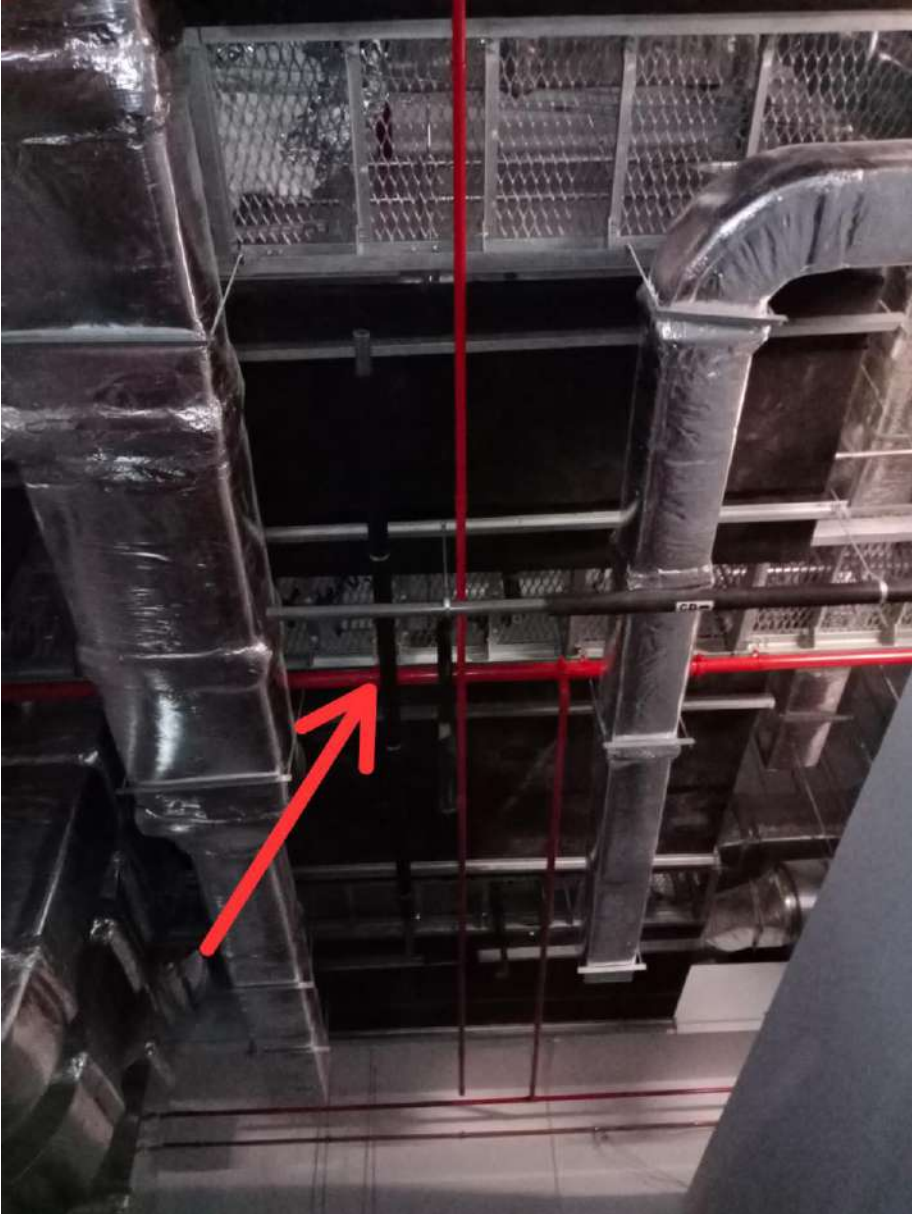


❖ Maintenance Space Awareness, Access Panel



Mechanical Ventilation Fans

Maintenance Space Awareness, Access Panel



Air Handling Unit

-Mount on the Floor

-Mount from Slab (less than <5000 kg)

Photo describe AHU hang from slab with accessible platform

Maintenance Space Awareness, Access Panel

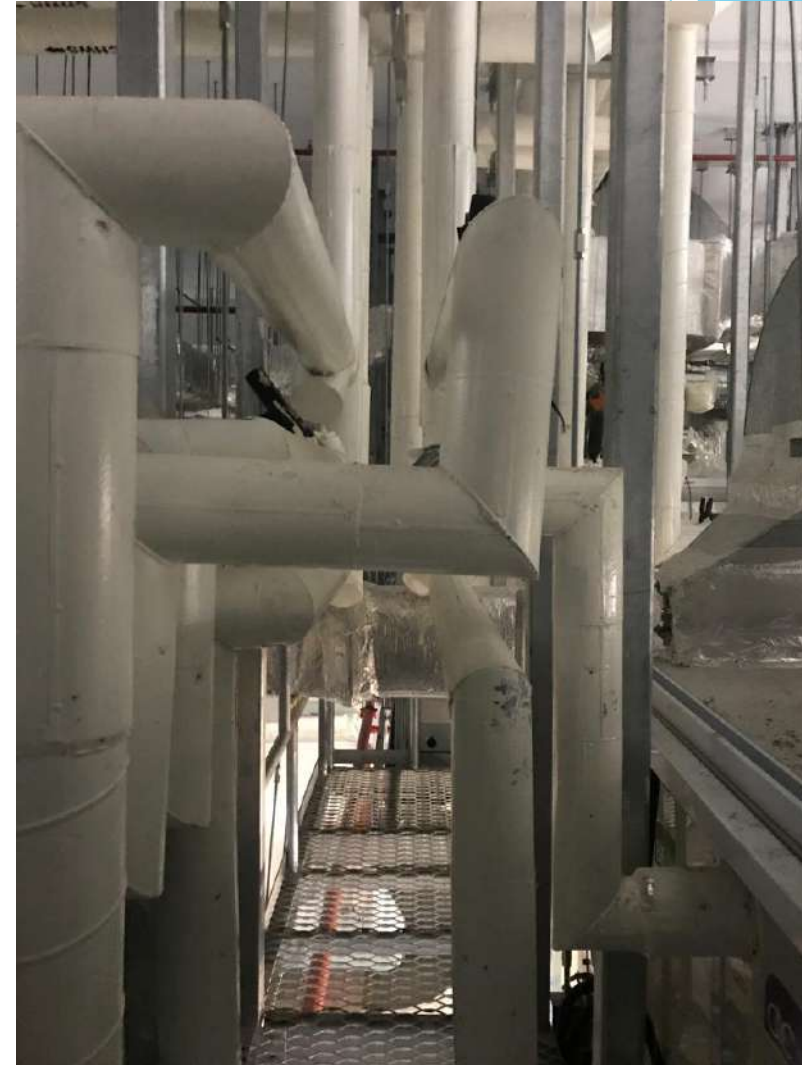


Photo describe AHU hang from slab with accessible platform

Maintenance Space Awareness, Access Panel

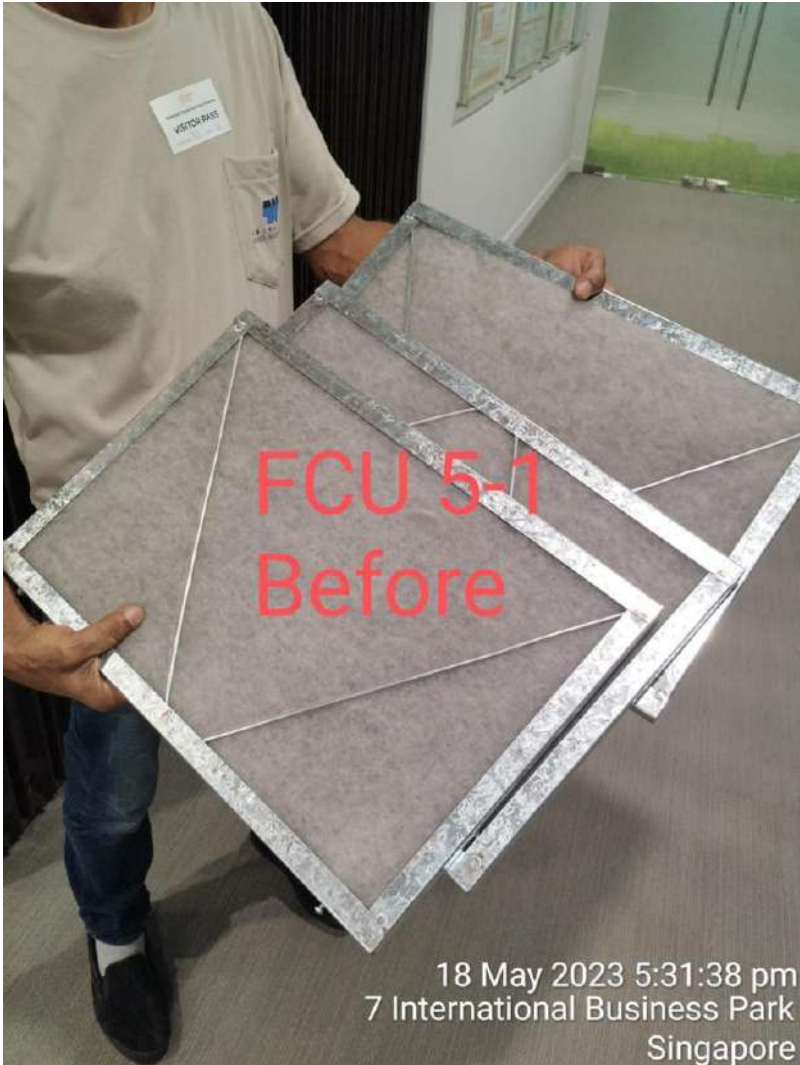


Photo describe Disposable Filter.

Maintenance Space Awareness, Access Panel



Photo describe the access panel on plaster ceiling.

Maintenance Space Awareness, Access Panel



Access Panel is blocking with Condensate Drin Pipe.



Filter Section is blocking with rods.
Too many electrical trunking run below the FCU.

Maintenance Space Awareness, Access Panel



Photo describe maintenance for Y-Strainer.

Maintenance Space Awareness, Access Panel

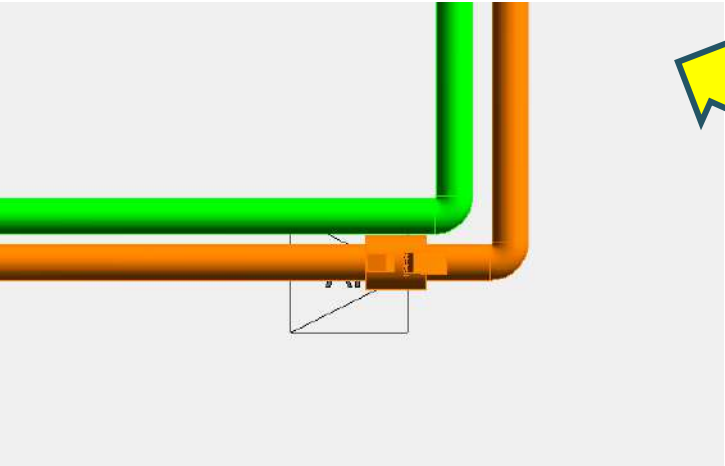


Access Panel is blocking with Condensate Drin Pipe.

Maintenance Space Awareness, Access Panel

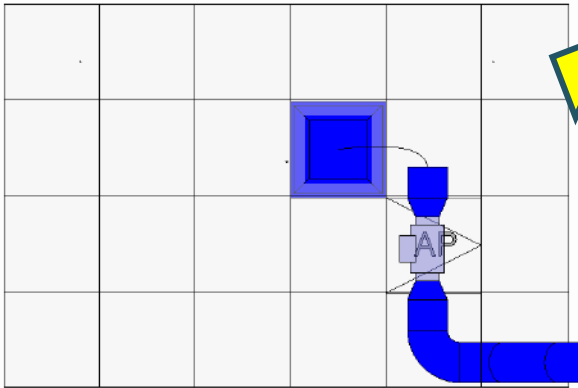
Fire Damper, Duct Heater, Volume Control Damper, Non-Return Damper, VAV Box, Local Control Panel

Request Access Panel for Valves



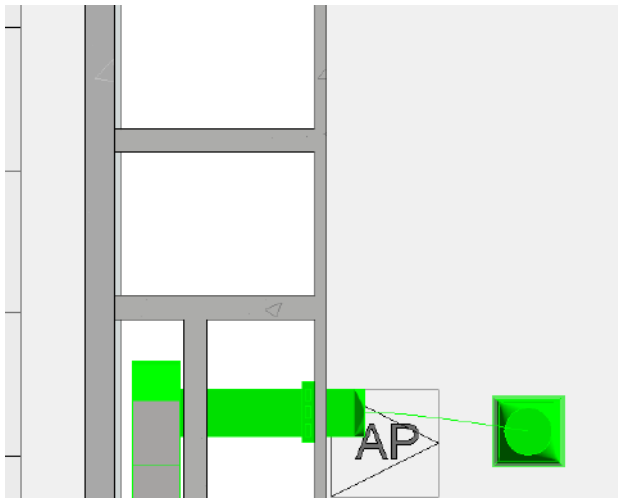
GYPSUM Ceiling

Request Access Panel for VAV Box



Grid Ceiling Type

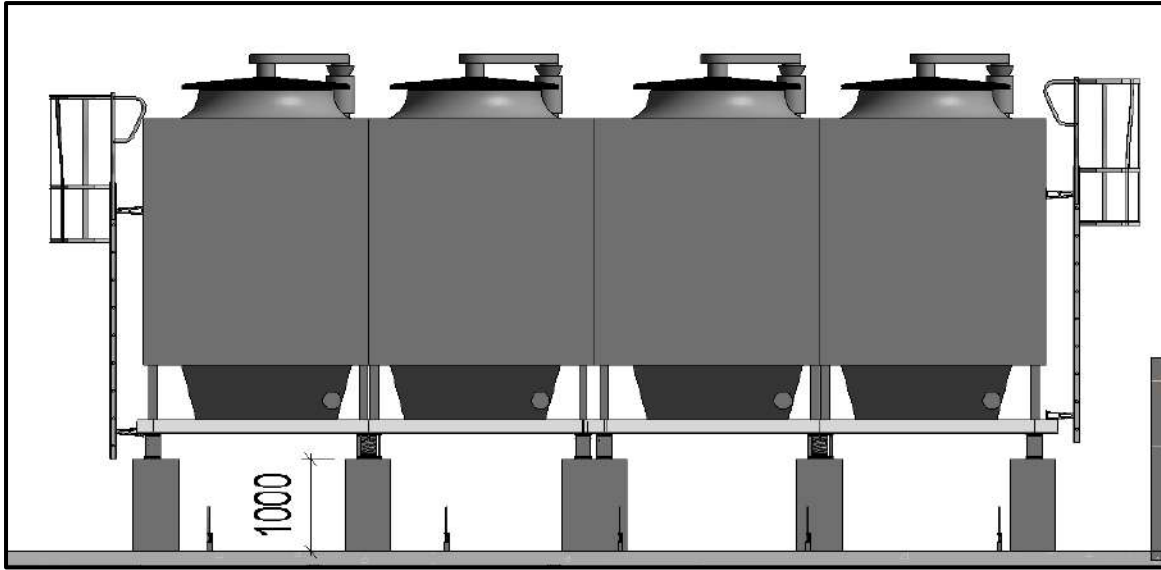
Request Access Panel for Fire Damper



Example: Type of Ceiling

GYPSUM CEILING IN PAINT FINISH
2HR FIRE RATED CALCIUM SILICATE BOARD IN EMULSION PAINT FINISH
600X1200MM PERFORATED ANTIBACTERIAL ACOUSTIC GRID CEILING
CALCIUM SILICATE BOARD CEILING IN EMULSION PAINT FINISH
600X600MM CALCIUM SILICATE GRID CEILING IN EMULSION PAINT FINISH
600X600MM ANTIBACTERIAL ACOUSTIC GRID CEILING
600X600MM ANTIBACTERIAL ACOUSTIC GRID CEILING WITH METAL CLIP DOWN
600X600MM CLEAN ROOM ANTIBACTERIAL ACOUSTIC GRID CEILING
HIGH DENSITY GLASS FIBRE ACOUSTIC CEILING BOARD

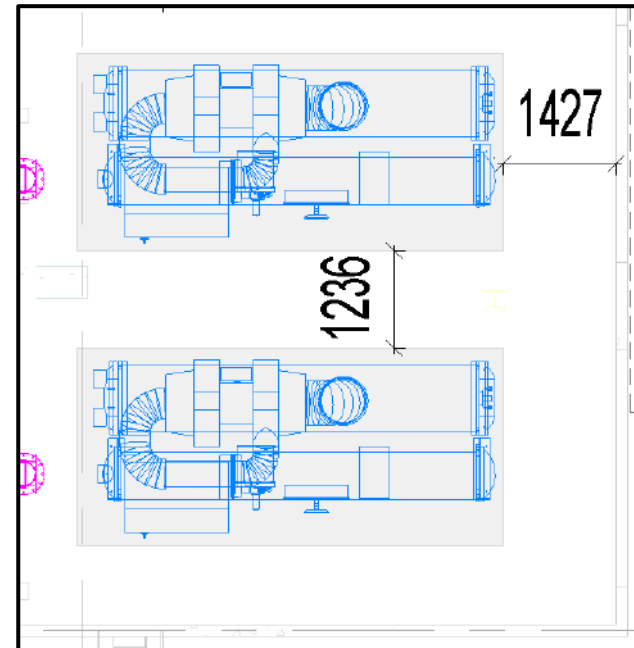
Maintenance Space Awareness, Access Panel



Chiller

Cooling Tower

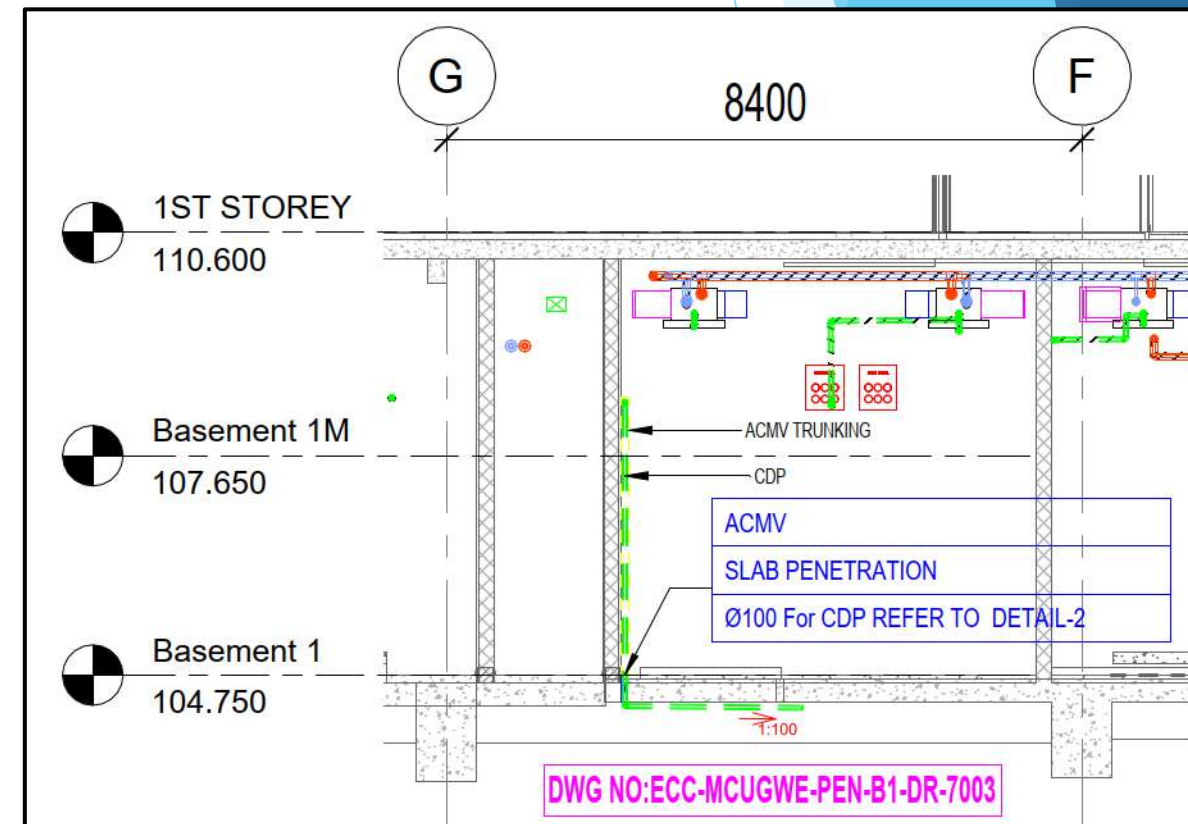
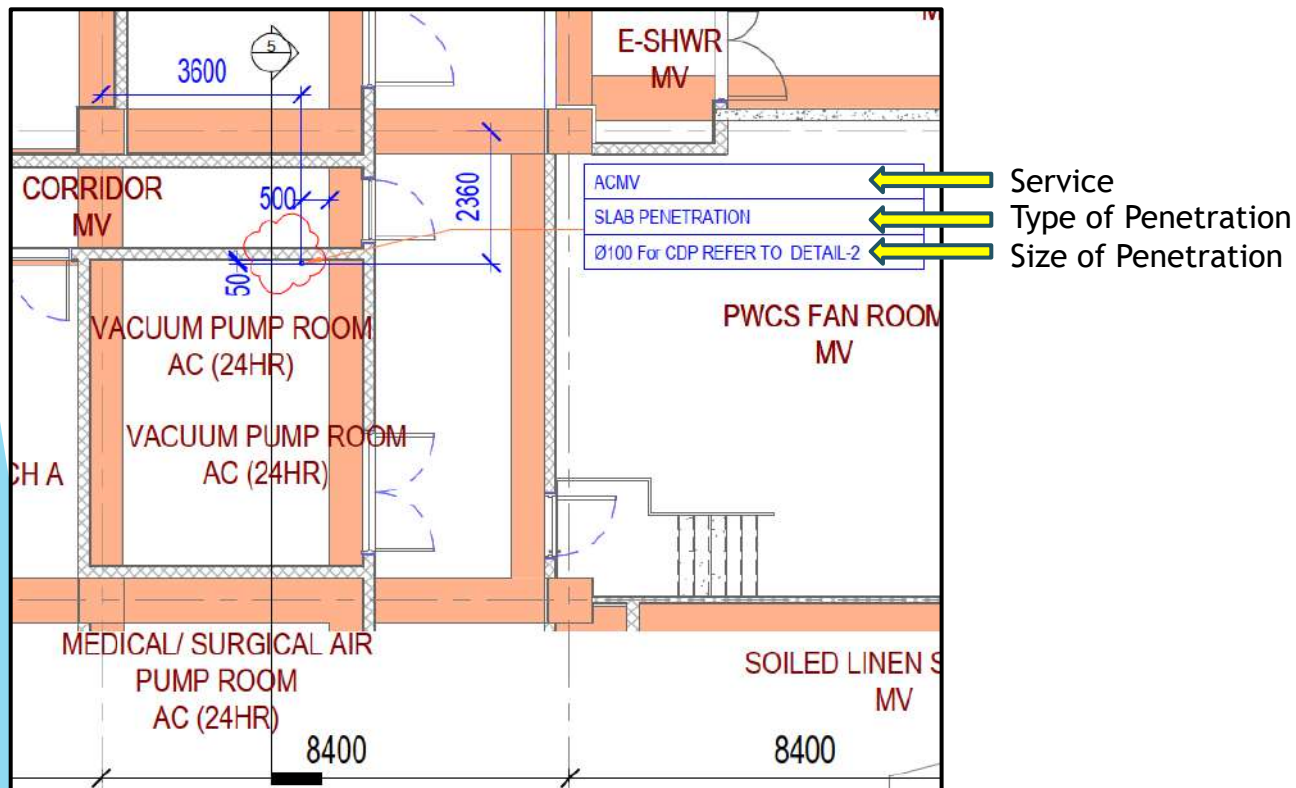
- If any other equipments need to accommodate like AHU, etc have to keep 5meter distance
- Need to consider High of Cooling Tower is not exceed to Building AMSL (Above Mean Sea Level)



❖ Opening Requirement

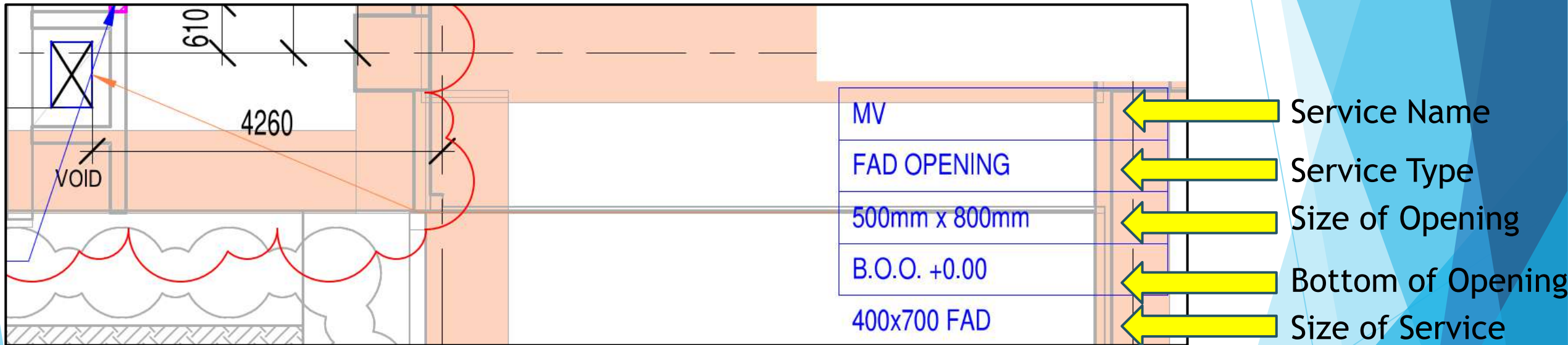
- ▶ We need to request opening when duct passes through the wall, slab, ceiling and RC concrete.
- ▶ There are three type of openings wall, floor, ceiling and RC concrete.

Plan Vs Section View for Condensate Drain Pipe



❖ Opening Requirement

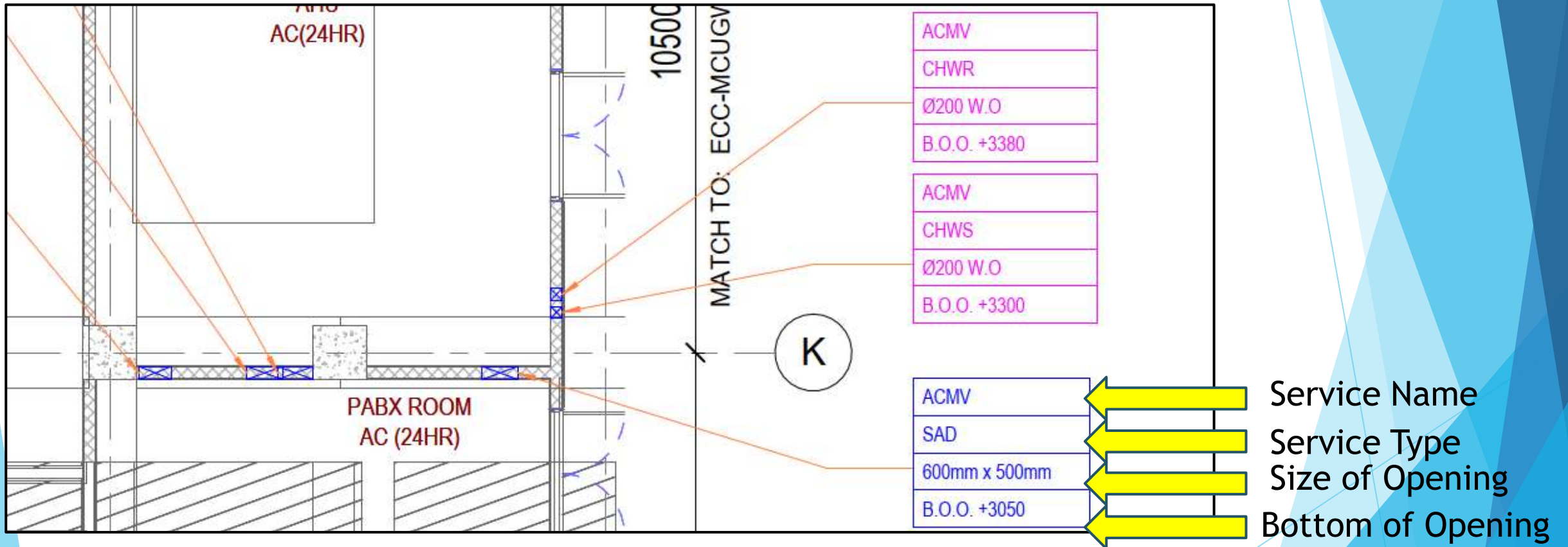
Plan View for Slab Penetration



The requirement of the opening size exceed round 50mm than the duct size.

❖ Opening Requirement

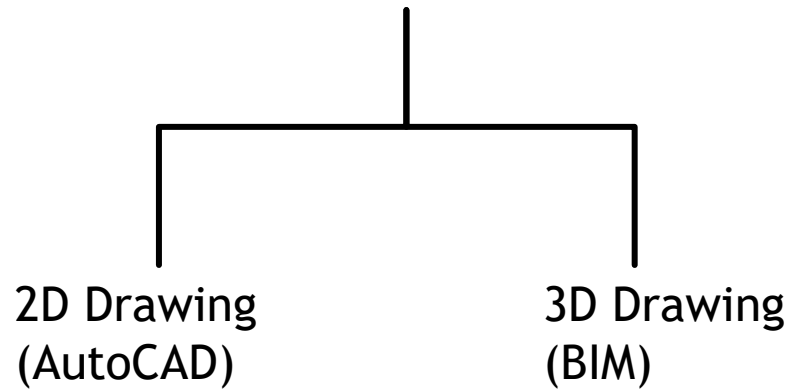
Plan View for Wall Penetration



The requirement of the opening size exceed round 50mm than the duct size.

Types of Drawings

- ▶ Tender Drawings (Archi, Structure, M&E, Schematic) >>> From Consultants
- ▶ ACMV Shop Drawings >>> From Sub-Contractor



- ▶ Combine Service Drawing >>> From Main-Contractor
- ▶ Room Book Drawing >>> input from individual sub-contractor system >>> final combine drawing from Main Contractor
- ▶ ACMV As-Built Drawing >>> After installation done on site

Usage of Color, Line Type, Text Style, Text Size, Scale and Paper Size

Before we print out the drawings the following steps need to be consider.....

- ✓ Use main color
- ✓ to adjust the line type
- ✓ to adjust font style and font size
- ✓ to set standardize scale on every plan drawings
- ✓ Legend should be standardize
- ✓ Drawing naming standard
- ✓ Use proper template
- ✓ Folder Structure
- ✓ AutoCAD >>> follow CP83
- ✓ Revit >>> follow BIM Execution Plan (BEP), BIM Quality Checklist

What is Room Book Drawing?

also known as Room Data Sheets or Room Finish Schedules, are architectural drawings that provide detailed information about each room in a building

❖ Room book drawings typically include the following information:

- Room Identification
- Dimensions
- Room Layout
- Wall Finishes
- Floor Finishes
- Ceiling Finishes
- Lighting Fixtures
- Electrical Outlets
- ACMV Systems
- Special Features (unique features or specialized equipment specific to a particular room)

ACC

B1.ACC.147	4.99 m ²	M2
FFL +104.750	SFL +104.650	
WALL FIN WP	WT-02	TYPE 2
SKETING	NA	
CL FIN WP	FT-01/FT-04	TYPE 2
CF LH	CB-04/CB-05	3000

21

SAN

B1.MEP.323	1.61 m ²	
FFL +104.750	SFL +104.700	
WALL FIN WP	WP-02	TYPE 2
SKETING	NA	
CL FIN WP	FC-05	TYPE 2
CF LH	CB-01	

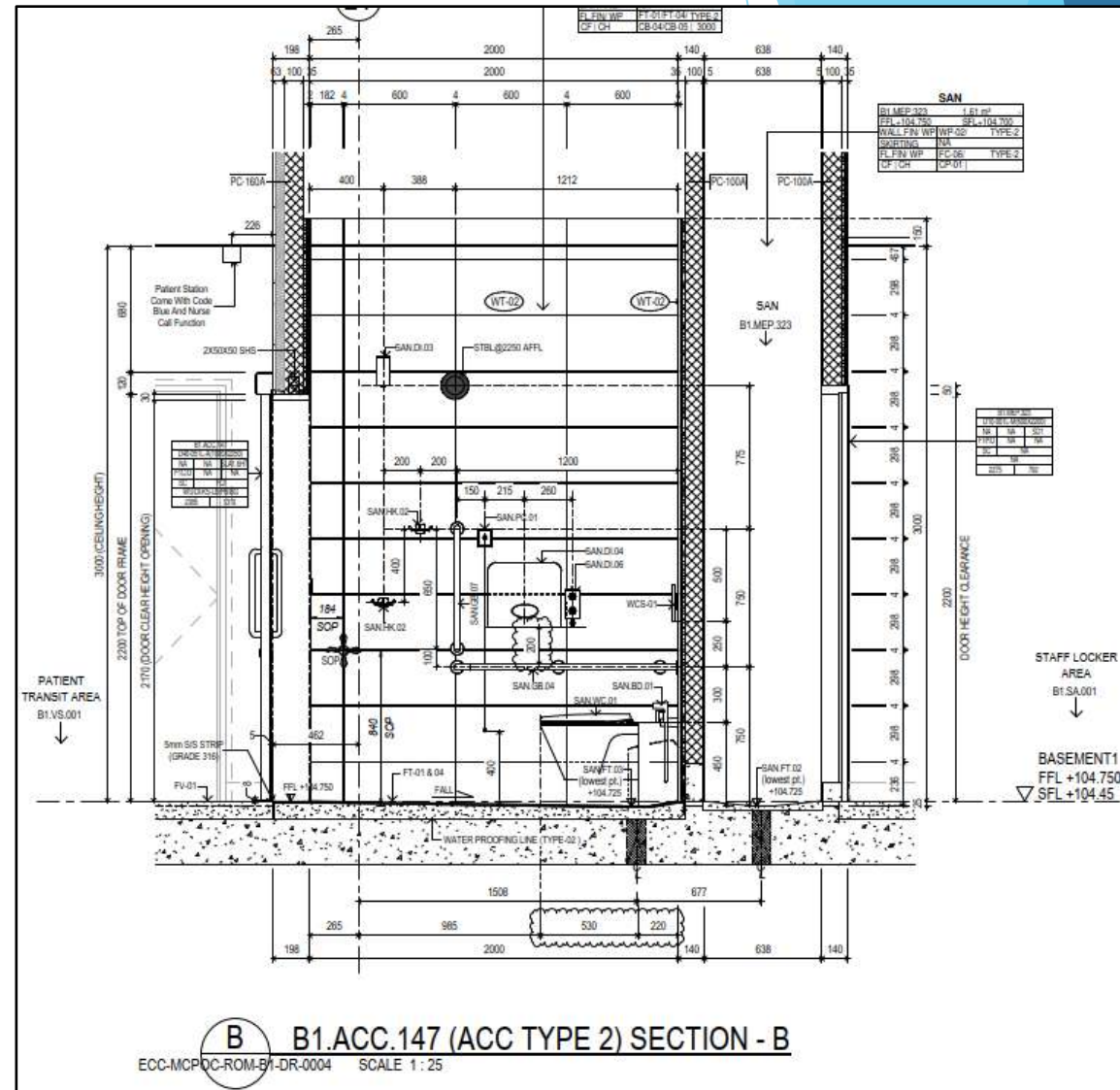
STAFF LOCKER AREA
B1.SA.001

PATIENT TRANSIT AREA
B1.VS.001

BASEMENT 1
FFL +104.750
SFL +104.45

B1.ACC.147 (ACC TYPE 2) SECTION - A

ECC-MCP-PC-ROM-B1-DR-0004 SCALE 1:25



thank you